

Evaluating the health of City of Toronto's street trees by taxonomy and size

INF2167 - R for Data Science

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Abstract

Introduction

Relevant Literature

Dataset & Methods

Dataset Source

The street tree dataset used for this project is sourced from Toronto Open Data (City of Toronto 2026), a publicly available data initiative maintained by the City of Toronto. This dataset was published by City of Toronto's Parks, Forestry, and Recreation division initially on December 1, 2011 and was last updated on June 4, 2026.

The original dataset contains 16 variables, focusing on unique IDs, geographic location, tree type, and tree size. There is a total of 688,335 rows in the dataset.

Data Cleaning

A new dataset variable was created, called `tree_data`. This dataset only contains variables that were necessary for this project, including:

- `OBJECTID`, a unique ID of each tree;
- `WARD`, the tree's location by geographical district;
- `BOTANICAL_NAME`, the tree's formal scientific name; and
- `DBH_TRUNK`, the tree's diameter at breast height.

Another new dataset was created, called `tree_data_clean`. The `janitor` package (Firke 2023) was used to clean up the variables names in the dataset so that names were in a consistent snake case format. A new “region” variable was created based on the existing “ward” variable by categorizing wards by region. For example, the trees in Wards 6, 8, 15, 16, 17, and 18 were categorized as being located in North York.

One of the focuses for this project is the 10-20-30 rule which requires the classification of the trees by species, genus, and family. Though species and genus may be inferred for some trees based on botanical name (e.g., if the botanical name is *Acer Platanoides*, the species is *Acer Platanoides* and the genus is *Acer*), the tree’s family name must be identified using an external source. Thus, the World Flora Online (WFO) package [WorldFlora2020] was used to create the variables, `species_name`, `genus_name`, and `family_name`, based on the existing variable, botanical name. To optimize runtime, a new variable, called `unique_trees`, was created to identify all unique botanical names in the `tree_data_clean` dataset. This table had a total of 304 unique tree names. By running the WFO package’s classification function on the `unique_trees` table instead of the `tree_data_clean` dataset, the function had to match 304 unique botanical names instead of matching names for each of the 688,335 tree entries individually, saving processing power and optimizing time efficiency.

After the botanical names in `unique_trees` were classified (as the variable, `taxonomy_extract`), new variables in the `tree_data_clean` dataset were created for `species_name`, `genus_name`, and `family_name`. The `left_join()` function (Wickham et al. 2019) was used to join the species, genus, and family names by the `botanical_name` variable present in both datasets. The results of the WFO matching were checked for missing values, and there were a total of 18 botanical names that the classification step was unable to classify.

botanical_name	n
<i>Acer x freemanii</i> (<i>A. rubrum x saccharinum</i>) 'Autumn Blaze'	15025
<i>Quercus alba</i>	3807
<i>Acer x freemanii</i> (<i>A. rubrum x saccharinum</i>) 'Armstrong'	2781
<i>Magnolia x soulangeana</i> (<i>M. denudata x liliiflora</i>)	2688
<i>Acer x freemanii</i> (<i>A. rubrum x saccharinum</i>) 'Marmo'	1000
<i>Acer x freemanii</i> (<i>A. rubrum x saccharinum</i>) 'Celebration'	965
<i>Acer x freemanii</i> (<i>A. rubrum x saccharinum</i>) 'Sienna Glen'	837
<i>Acer x freemanii</i> (<i>A. rubrum x saccharinum</i>)	563
<i>Salix x sepulcralis</i> 'Chrysocoma'	545
<i>Ulmus x hollandica</i> (<i>U. glabra x minor</i>) 'Pioneer'	425
<i>Populus x canadensis</i> (<i>Populus nigra x deltoides</i>)	249
<i>Tilia x euchlora</i> (<i>T. cordata x dasystyla</i>)	151
<i>Prunus x cistena</i> (<i>P. pumila x P. cerasifera</i> 'Atropurpurea')	133
<i>alix x sepulcralis</i>	101
None	56
<i>Prunus x yedoensis</i> 'unknown hybrid'	31
<i>Amelanchier x grandiflora</i> (<i>A. arborea x laevis</i>) 'Robin Hill'	15
<i>Sorbus aria</i>	4

The missing data was resolved manually by identifying the species, genus, and family names for botanical names that appear more than 1000 times in the tree_data_clean dataset. Missing classifications for botanical names with less than 1000 trees were considered insignificant. Manual identification was used for *Acer x freemanii*, *Quercus alba*, and *Magnolia x soulangeana*. *Acer x freemanii* made up a significant portion of Toronto's street tree population with a total of 21,171 trees.

Methods

Exploratory data analysis was conducted to evaluate ward-level compliance to the 10-20-30 rule. Linear regression modelling was then used to investigate the relationship between the 10-20-30 rule compliance and trunk size (DBH) distributions in wards.

Results & Findings

Exploratory Data Analysis

ward	ward_name	max_species	dominant_species	max_genus	dominant_genus	max_family	dominant_family
1	Etobicoke North	0.1635295	Acer platanoides	0.2383636	Acer	0.2589396	Sapindaceae
2	Etobicoke Centre	0.1825712	Acer platanoides	0.2985332	Acer	0.3193174	Sapindaceae
3	Etobicoke-Lakeshore	0.1328476	Acer platanoides	0.2896189	Acer	0.3067255	Sapindaceae
4	Parkdale-High Park	0.1199773	Acer platanoides	0.2569476	Acer	0.2715687	Sapindaceae
5	York South-Weston	0.1276188	Acer platanoides	0.2645255	Acer	0.2817541	Sapindaceae
6	York Centre	0.1131337	Acer platanoides	0.2096812	Acer	0.2277967	Sapindaceae
7	Humber River-Black Creek	0.1054473	Acer platanoides	0.1820105	Acer	0.2014564	Sapindaceae
8	Eglinton-Lawrence	0.1418734	Acer platanoides	0.2719099	Acer	0.2861086	Sapindaceae
9	Davenport	0.1148122	Acer platanoides	0.2606241	Acer	0.2724800	Sapindaceae
10	Spadina-Fort York	0.1252920	Gleditsia triacanthos	0.2145020	Acer	0.2347495	Sapindaceae
11	University-Rosedale	0.1323976	Acer platanoides	0.2868825	Acer	0.3042910	Sapindaceae

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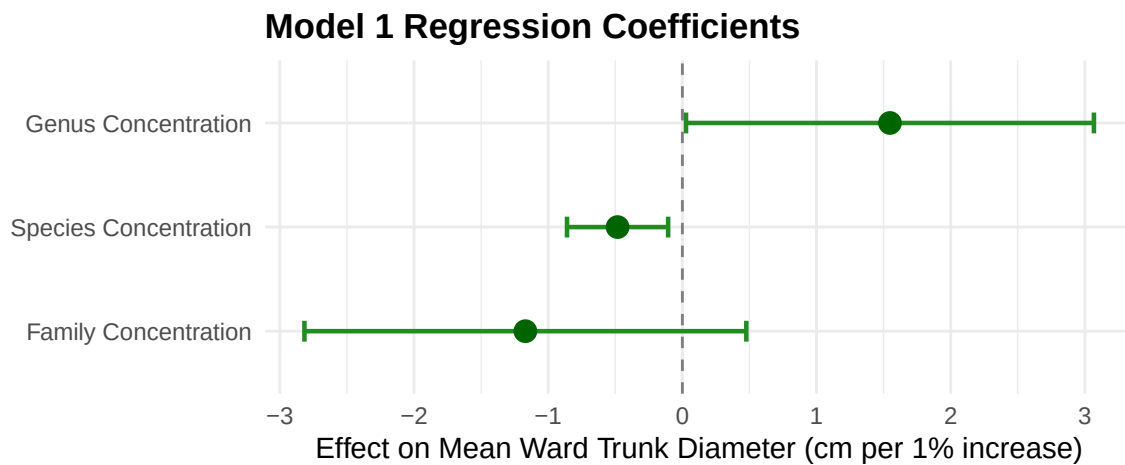
ward	ward_name	max_species	domi- nant_species	max_genus	domi- nant_genus	max_family	domi- nant_family
12	Toronto-St. Paul's	0.1634760	Acer platanoides	0.2879129	Acer	0.2976351	Sapindaceae
13	Toronto Centre	0.1648532	Gleditsia triacanthos	0.1945712	Gleditsia	0.2411372	Fabaceae
14	Toronto- Danforth	0.1456086	Acer platanoides	0.3193180	Acer	0.3362101	Sapindaceae
15	Don Valley West	0.1701510	Acer platanoides	0.2906350	Acer	0.2998805	Sapindaceae
16	Don Valley East	0.1530587	Acer platanoides	0.2586868	Acer	0.2688010	Sapindaceae
17	Don Valley North	0.1736035	Acer platanoides	0.2707713	Acer	0.2842581	Sapindaceae
18	Willowdale	0.1373216	Acer platanoides	0.2731174	Acer	0.2885025	Sapindaceae
19	Beaches- East York	0.1264929	Acer platanoides	0.2830332	Acer	0.2965877	Sapindaceae
20	Scarbor- ough Southwest	0.1154172	Acer platanoides	0.2658264	Acer	0.2853447	Sapindaceae
21	Scarbor- ough Centre	0.1250248	Acer platanoides	0.2531231	Acer	0.2755800	Sapindaceae
22	Scarborough- Agincourt	0.2018939	Acer platanoides	0.3152064	Acer	0.3413165	Sapindaceae
23	Scarbor- ough North	0.1816069	Acer platanoides	0.2976157	Acer	0.3317171	Sapindaceae
24	Scarborough- Guildwood	0.1434368	Acer platanoides	0.2731563	Acer	0.3005028	Sapindaceae
25	Scarborough- Rouge Park	0.1634354	Acer platanoides	0.3169567	Acer	0.3351326	Sapindaceae
*							

ward	ward_name	total_trees_ward	mean_dbh	median_dbh	sd_dbh
1	Etobicoke North	37082	20.31465	11	19.55837
2	Etobicoke Centre	52155	27.62345	23	22.83915
3	Etobicoke-Lakeshore	46941	27.20762	19	30.76771
4	Parkdale-High Park	28247	28.94775	20	27.34940
5	York South-Weston	28209	24.37513	18	22.09037
6	York Centre	34004	21.17298	14	18.80286
7	Humber River-Black Creek	24581	17.79952	10	16.61476

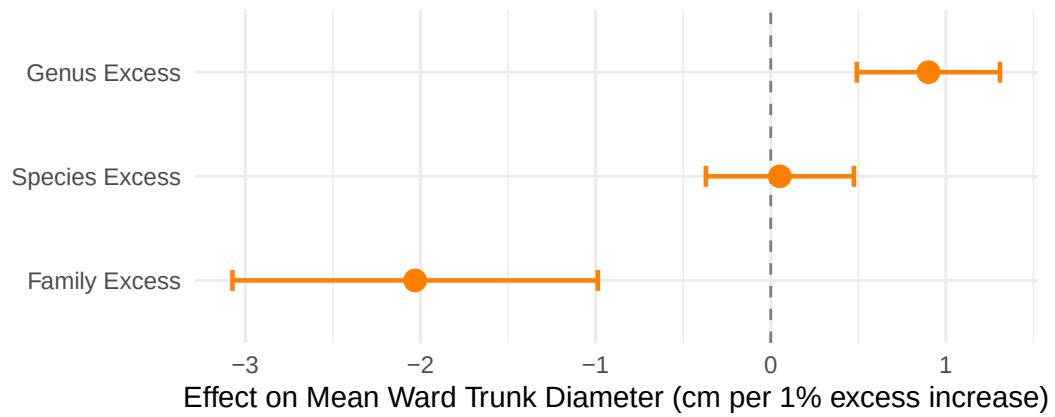
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ward	ward_name	total_trees_ward	mean_dbh	median_dbh	sd_dbh
8	Eglinton-Lawrence	35144	26.94986	20	25.35631
9	Davenport	19484	23.25549	15	22.25355
10	Spadina-Fort York	11557	21.57177	14	19.58469
11	University-Rosedale	23724	25.31011	16	24.05542
12	Toronto-St. Paul's	26640	26.39790	18	25.25155
13	Toronto Centre	8547	22.46098	15	19.69410
14	Toronto-Danforth	19003	27.42788	20	26.31245
15	Don Valley West	40993	30.17276	25	51.63810
16	Don Valley East	24520	25.77276	20	19.84693
17	Don Valley North	31883	25.91187	22	20.14952
18	Willowdale	31459	27.69465	23	22.49197
19	Beaches-East York	21100	27.42261	20	26.03292
20	Scarborough Southwest	21262	24.16758	13	24.46260
21	Scarborough Centre	20172	21.00570	8	21.51679
22	Scarborough-Agincourt	21754	20.83654	12	19.76509
23	Scarborough North	24955	18.34847	10	15.04367
24	Scarborough-Guildwood	22672	21.61547	10	21.46700
25	Scarborough-Rouge Park	27234	20.74007	14	18.59513
*					

Linear Regression Modelling



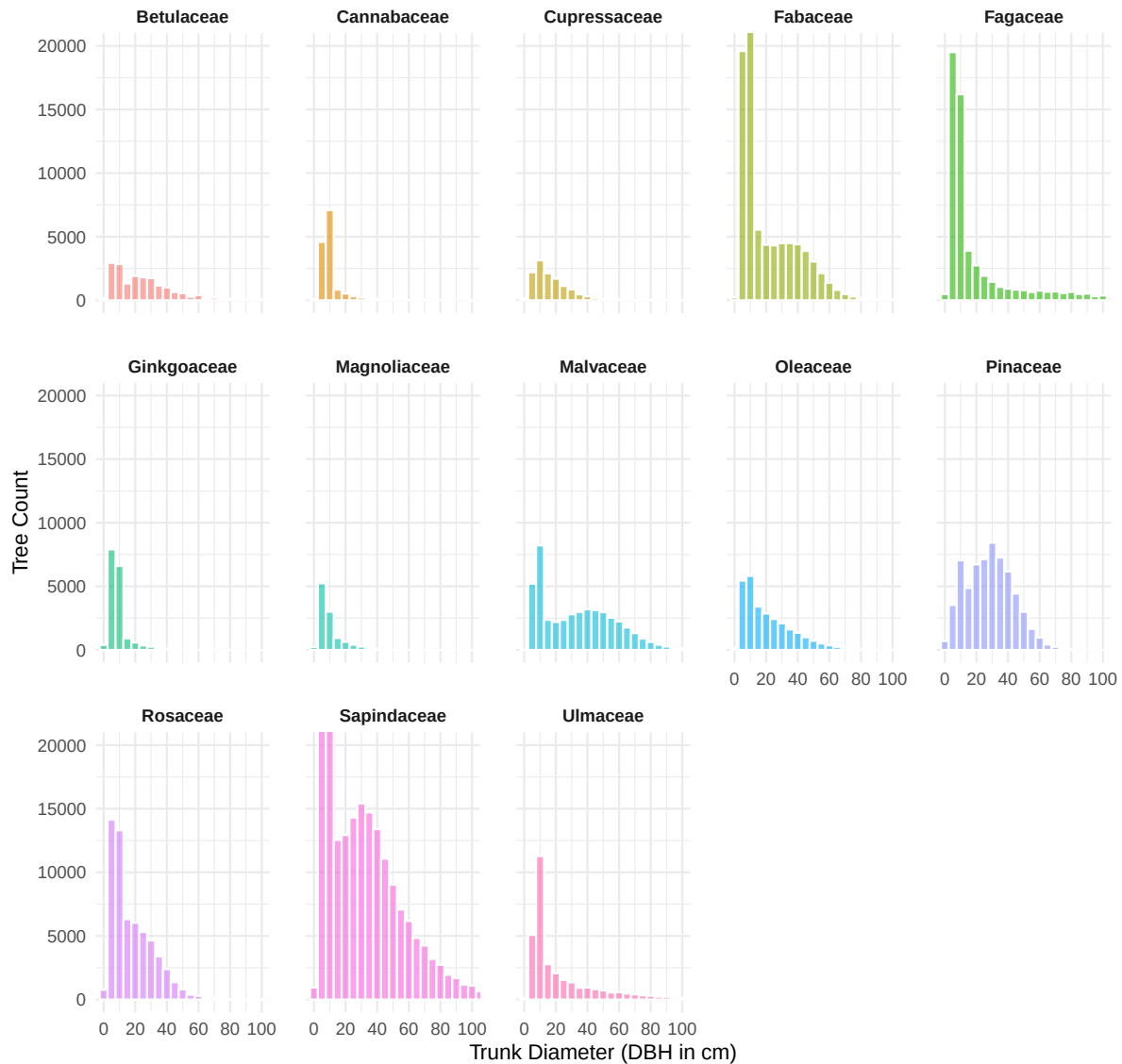
Model 2 Regression Coefficients



Trunk Diameter Distribution Across Top Genus Types



Trunk Diameter Distribution Across Top Tree Families



Discussion & Conclusion

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